Reg. No.														

B.Tech. DEGREE EXAMINATION, NOVEMBER 2019
Third Semester

18BTC101J – BIOCHEMISTRY
(For the candidates admitted during the academic year 2018-2019 onwards)

Note:
(i) **Part - A** should be answered in OMR sheet within first 45 minutes and OMR sheet should be handed over to hall invigilator at the end of 45th minute.
(ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: Three Hours — Max. Marks: 100

PART – A (20 × 1 = 20 Marks)
Answer ALL Questions

1. A Lewis base is a
(A) Lone pair acceptor (B) Lone pair donor
(C) Proton acceptor (D) Proton donor

2. An example for essential fatty acid is
(A) Linoleic acid (B) Palmitic acid
(C) Acetic acid (D) Myrstic acid

3. Boat and chair confirmations are found in
(A) Pyranose sugar (B) Furanose sugar
(C) All the sugars (D) Only D sugars

4. How many amino acids are present in glutathione?
(A) 2 (B) 4
(C) 3 (D) 5

5. The final output of the Krebs cycle includes all of the following EXCEPT
(A) NADP (B) $FADH_2$
(C) ATP (D) CO_2

6. In liver, glycogenolysis is negatively regulated by
(A) Glucagon (B) Epinephrine
(C) Insulin (D) Dopamine

7. How the pancreatic β-cells trigger the insulin secretion?
(A) An increase blood glucose (B) Decrease the blood glucose
(C) Increase an albumin in urine (D) No changes in blood glucose

8. Which of the following is a multi enzyme complex?
(A) Glyceraldehyde-3-phosphatase (B) Isomerase
(C) Pyruvate dehydrogenase (D) Hexokinase

9. Which of the following is a characteristic of phenylketonuria?
   (A) Deficiency of phenyl alanine hydroxylase
   (B) α–ketoglutarate defect
   (C) Abudance of reductase
   (D) High levels of phenyl alanine hydroxylase

10. Which of the following amino acid can covalently bind to ammonia, transport it and store it in a non-toxic form?
   (A) Aspartate
   (B) Glutamate
   (C) Serine
   (D) Cysteine

11. Transaminase enzymes are present in
   (A) Liver
   (B) Pancreas
   (C) Intestine
   (D) Muscle

12. Which of the following compounds is NOT synthesized by tryptophan?
   (A) Serotonin
   (B) Melatonin
   (C) Niacin
   (D) Epinephrine

13. Each cycle of β-oxidation produces
   (A) 1 FAD, 1 NADH and 1 acetyl – CoA
   (B) 1 $FADH_2$, 1 NADH and 1 acetyl CoA
   (C) 1 FADH, 1 NAD and 2 CO_2
   (D) 1 $FADH_2$, 1 NADH and 2 CO_2

14. All the 27 carbon atoms of cholesterol are derived from
   (A) Acetyl CoA
   (B) Acetoacetyl CoA
   (C) Propionyl CoA
   (D) Malonyl CoA

15. A synthetic nucleotide analogue, 4-hydroxy-pyrozolo pyrimidine is used in the treatment of
   (A) Acute nephritis
   (B) Cystic fibrosis
   (C) Multiple myeloma
   (D) Gout

16. A pyrimidine nucleotide is
   (A) GMP
   (B) AMP
   (C) CMP
   (D) IMP

17. Which of the following statements about the malate-aspartate shuttle is false?
   (A) 3 ATP are formed per shuttle
   (B) Cytoplasmic oxaloacetate is converted to malate
   (C) Mitochondrial malate is converted to oxaloacetate
   (D) Anti-port transport is not involved

18. When electrons are removed from pyruvate in the transition reaction, they are accepted by
   (A) Acetyl CoA
   (B) FAD
   (C) NAD^+
   (D) ATP

19. In aerobic respiration, the compound that enters a mitochondrion is
   (A) Acetyl CoA
   (B) Pyruvate
   (C) Phospho glyceraldehyde
   (D) Oxaloacetate

20. During glycolysis, electrons removed from glucose are passed to
   (A) Pyruvic acid
   (B) Lactic acid
   (C) FAD
   (D) NAD

PART – B (5 × 4 = 20 Marks)
Answer **ANY FIVE** Questions

21. Write the chemical reactions of monosaccharides.

22. Describe about the action of enzymes.

23. What are anaplerotic reactions?

24. Explain transamination.

25. Write the biosynthesis of purine.

26. Explain any one of the shuttle pathway.

27. Write the pathophysiology of diabetes mellitus.

PART – C (5 × 12 = 60 Marks)
Answer **ALL** Questions

28. a. Describe the chemical properties of fats and oils.

(OR)

b. Explain the different structures of protein with suitable diagrams.

29. a. Describe the steps of glycolysis and its regulatory steps.

(OR)

b. With the scheme, explain the TCA cycle.

30. a. Write about urea cycle.

(OR)

b. Describe the amino acids metabolic disorders.

31. a. Explain the process through which pyrimidines are synthesized.

(OR)

b. Discuss about steps involved in β-oxidation of lipids.

32. a. Explain the chemisosmotic theory with suitable diagram.

(OR)

b. Describe the structure of electron transport system.

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